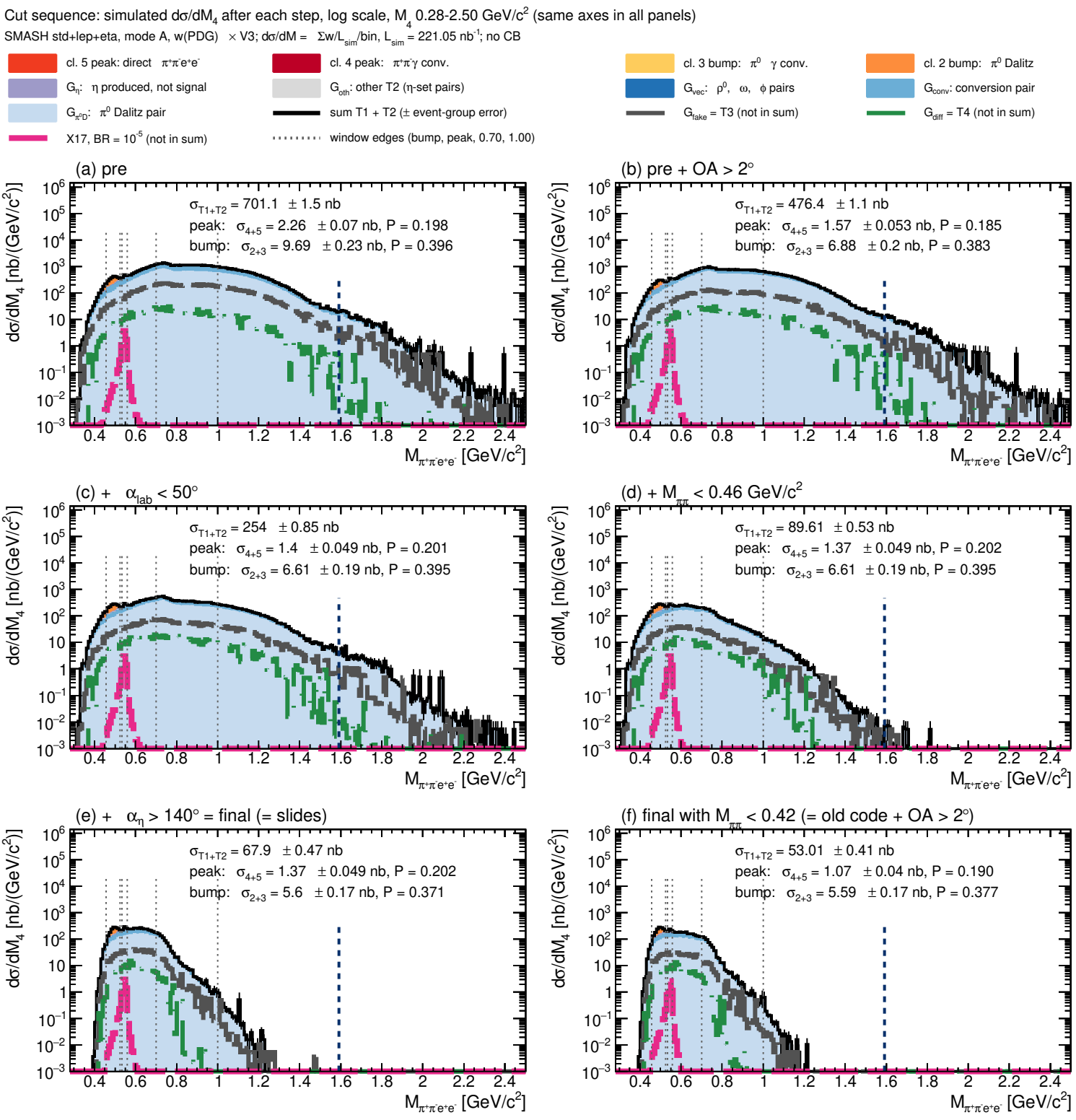

Cut sequence: simulated dσ/dM4 after each step, log scale, M4 0.28-2.50 GeV/c² (same axes in all panels)
SMASH std+lep+eta, mode A, w(PDG) × V3; dσ/dM = Σw/Lsim/bin, Lsim = 221.05 nb⁻¹; no CB
cl. 5 peak: direct π⁺π⁻e⁺e⁻
cl. 4 peak: π⁺π⁻γ conv.
cl. 3 bump: π⁰ γ conv.
cl. 2 bump: π⁰ Dalitz
Gη: η produced, not signal
Goth: other T2 (η-set pairs)
Gvec: ρ⁰, ω, ϕ pairs
Gconv: conversion pair
Gπ⁰D: π⁰ Dalitz pair
sum T1 + T2 (± event-group error)
Gfake = T3 (not in sum)
Gdiff = T4 (not in sum)
X17, BR = 10⁻⁵ (not in sum)
window edges (bump, peak, 0.70, 1.00)
(a) pre
σT1+T2 = 701.1 ± 1.5 nb
peak: σ4+5 = 2.26 ± 0.07 nb, P = 0.198
bump: σ2+3 = 9.69 ± 0.23 nb, P = 0.396
(b) pre + OA > 2°
σT1+T2 = 476.4 ± 1.1 nb
peak: σ4+5 = 1.57 ± 0.053 nb, P = 0.185
bump: σ2+3 = 6.88 ± 0.2 nb, P = 0.383
(c) + αlab < 50°
σT1+T2 = 254 ± 0.85 nb
peak: σ4+5 = 1.4 ± 0.049 nb, P = 0.201
bump: σ2+3 = 6.61 ± 0.19 nb, P = 0.395
(d) + Mππ < 0.46 GeV/c²
σT1+T2 = 89.61 ± 0.53 nb
peak: σ4+5 = 1.37 ± 0.049 nb, P = 0.202
bump: σ2+3 = 6.61 ± 0.19 nb, P = 0.395
(e) + αη > 140° = final (= slides)
σT1+T2 = 67.9 ± 0.47 nb
peak: σ4+5 = 1.37 ± 0.049 nb, P = 0.202
bump: σ2+3 = 5.6 ± 0.17 nb, P = 0.371
(f) final with Mππ < 0.42 (= old code + OA > 2°)
σT1+T2 = 53.01 ± 0.41 nb
peak: σ4+5 = 1.07 ± 0.04 nb, P = 0.190
bump: σ2+3 = 5.59 ± 0.17 nb, P = 0.377
dσ/dM4 [nb/(GeV/c²)]
Mπ⁺π⁻e⁺e⁻ [GeV/c²]